

Human-AI Interaction:

AI Transparency & Interpretability

Motahhare Eslami

Some Logistics

- **Homework 1**
 - feedback will be provided by Wed
- **This topic (Transparency and Interpretability) will be in 2 sessions** (today and next session)

INITIALLY

1
Make clear
what the
system
can do.

2
Make clear
how well the
system can
do what it
can do.

Guidelines for Human AI Interaction

Learn more: <https://aka.ms/aiguideelines>



DURING
INTERACTION

3
Time
services
based on
context.

4
Show
contextually
relevant
information.

5
Match
relevant
social norms.

6
Mitigate
social biases.

WHEN WRONG

7
Support
efficient
invocation.

8
Support
efficient
dismissal.

9
Support
efficient
correction.

10
Scope
services
when in
doubt.

11
Make clear
why the
system did
what it did.

OVER TIME

12
Remember
recent
interactions.

13
Learn from
user
behavior.

14
Update and
adapt
cautiously.

15
Encourage
granular
feedback.

16
Convey the
consequences
of user
actions.

17
Provide
global
controls.

18
Notify users
about
changes.

Trust + ethical issues are reported barriers to successful AI adoption

Of professionals surveyed

87% say bias in their data produce
discriminatory results ¹

65% can't explain how model
decisions or predictions are made ²

¹ [Alation State of Data Culture Q1 2021 Report](#)

² [FICO The State of Responsible AI: 2021](#)

- Transparency of Existence
- Transparency of Operation

Today

- Three explainable AI approaches
 - Feature attributions
 - Example-based explanations
 - Model analysis
- Two famous algorithms (for feature attribution) - SHAP and LIME
- Limitations of the explainable AI approaches
- In-class activities

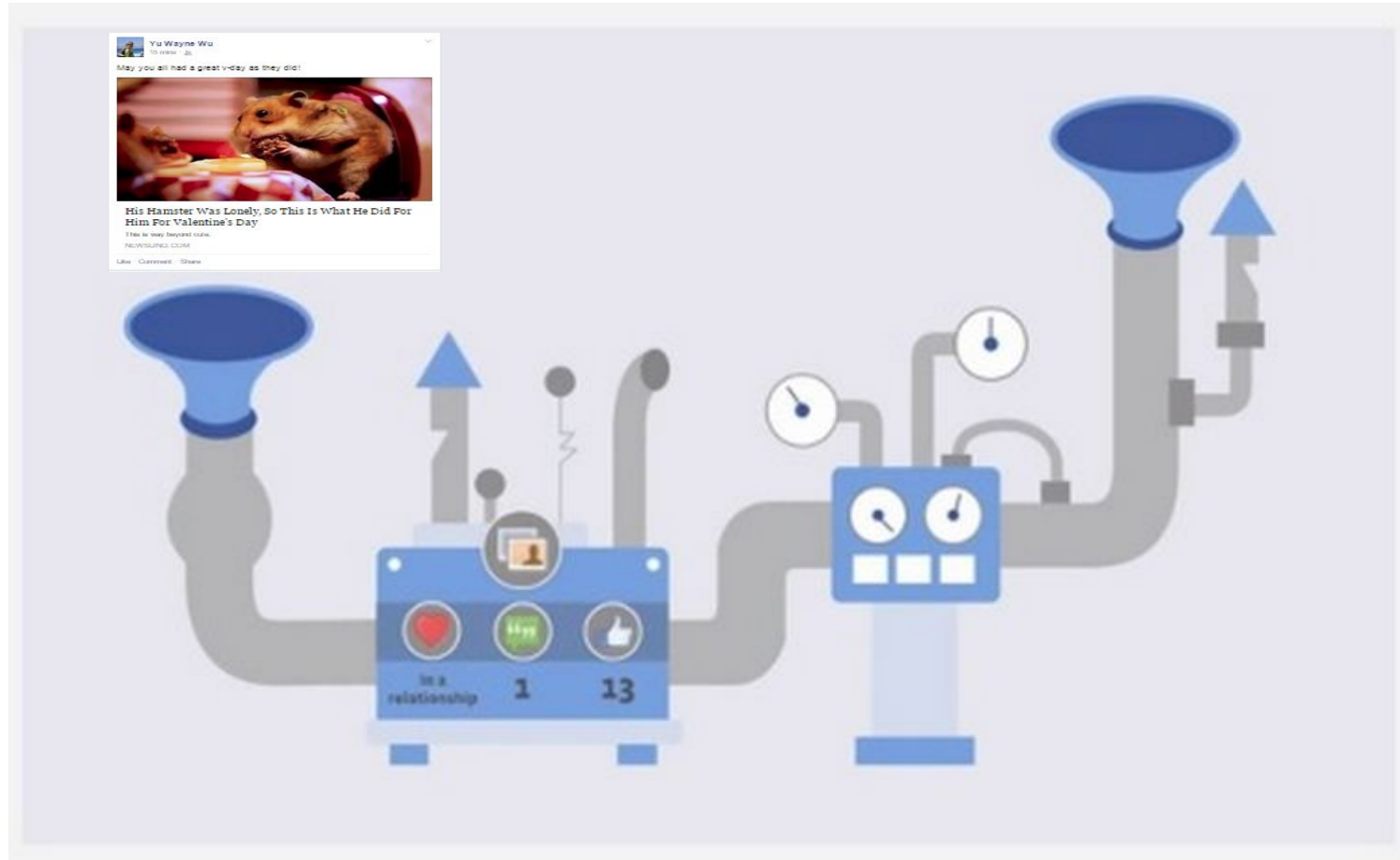
First things first:

Whether, and How are these different?

- Transparency
- Explainability (XAI)
- Interpretability (Interpretable ML)

- Transparency of Existence
- Transparency of Operation

Case Study: Facebook News Feed Filtering Algorithm



Score: 1.1

“Sarah” Scenario

- Your friend, “*Sarah*,” shared a public story on her wall visible to all her friends on Facebook.
- Would this story appear on your News Feed, if you make sure to check your news feed carefully?

Unaware (62.5%)
Aware (37.5%)

Unaware of the Filtering Algorithm's Existence

“I bet it would be on my news feed. I probably would catch [it] at some point during the day.” (P30)

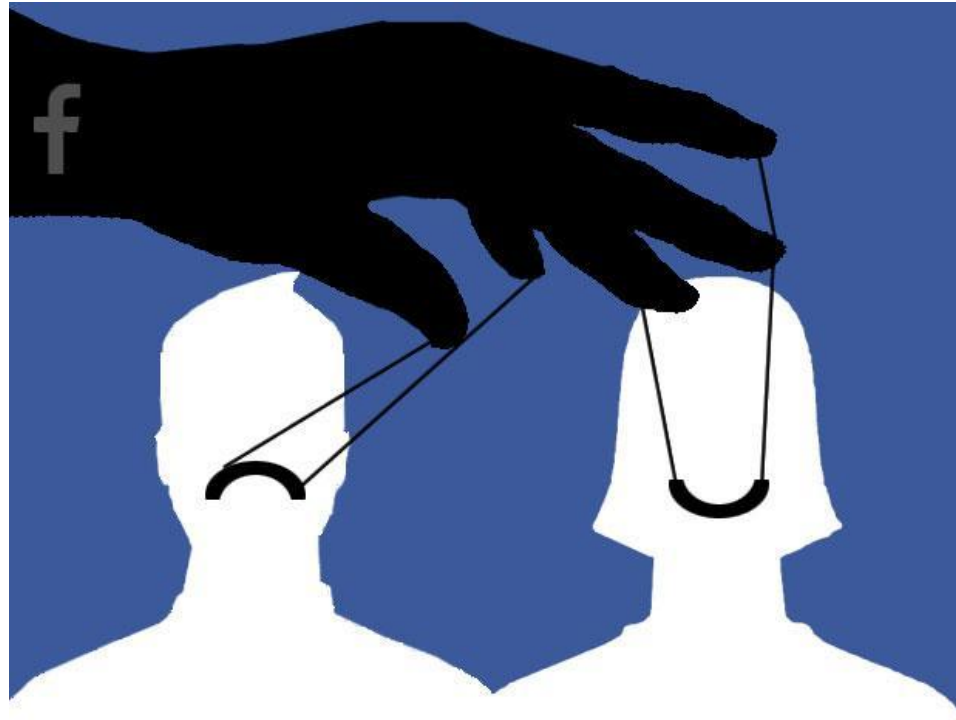
*“... and probably I don't scroll down enough but you know [you] cannot always scroll down so much, you just go like a few posts and that's it.”
(P9)*

“...and if I wanna go back to [a missed story], it's accessible” (P39)

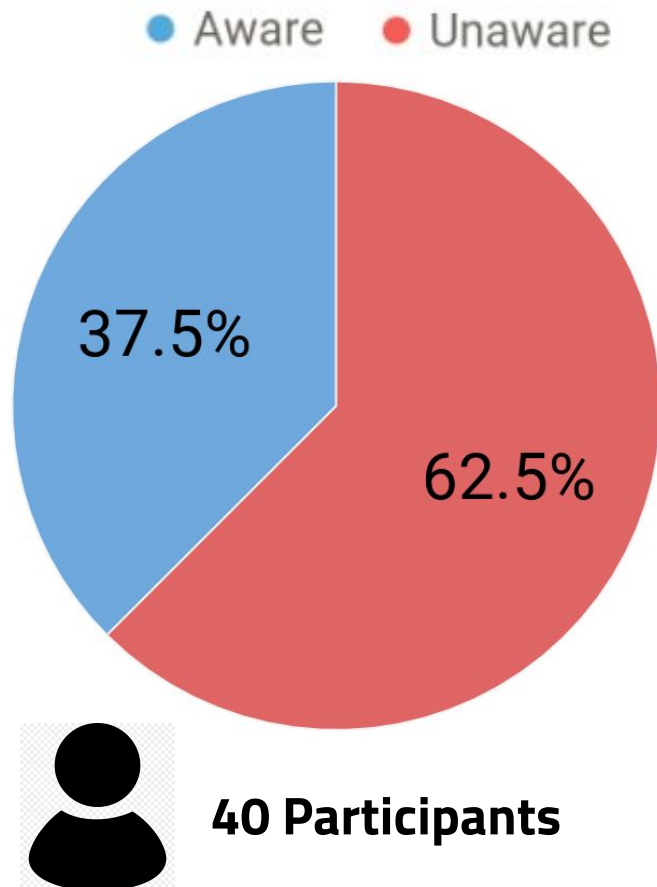
The Consequences of Unawareness

“ ‘How dare you manipulate my news feed!’...was a really fervent response—and very common.”

Jeff Hancock, Microsoft Faculty Summit, 2014



Lack of Awareness



Actively Engaged



More Prone to be Aware
of the Algorithm



Passively
Engaged

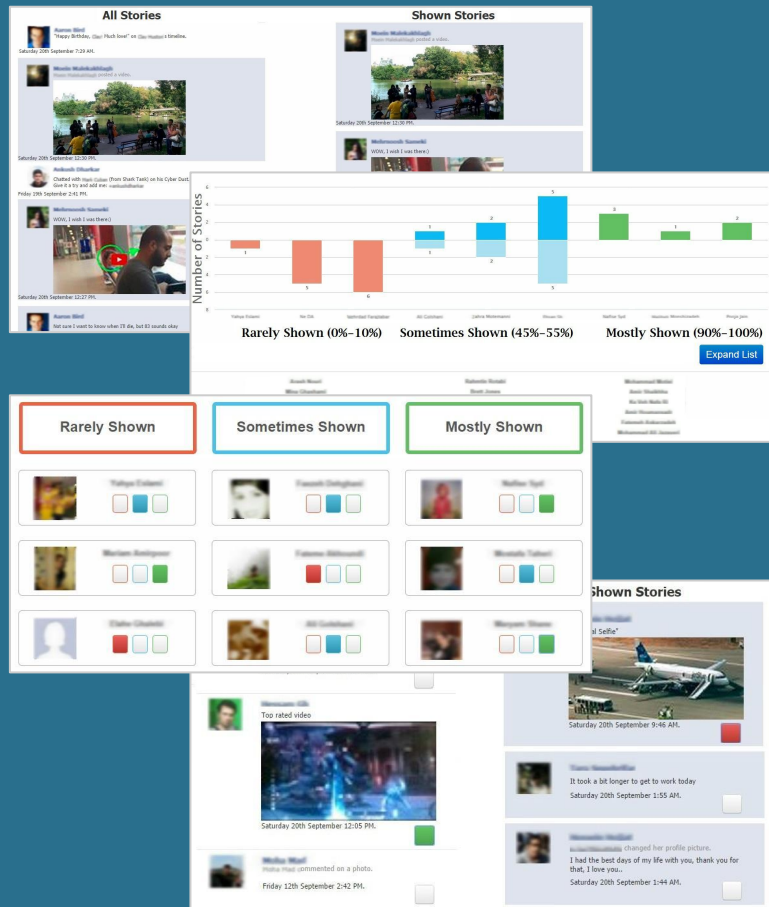


Less Prone to be Aware
of the Algorithm

FeedVis

The Algorithm Outputs

The differences in users' News Feeds when they have been curated by the algorithm and when they have not.



All Stories

 **Aaron Bird**
"Happy Birthday, Claire! Much love!" on Claire Hunter's timeline.
Saturday 20th September 7:29 AM.

 **Moeen Mahabubulhaghi**
Moeen Mahabubulhaghi posted a video.

Saturday 20th September 12:30 PM.

 **Anindita Dharwar**
Chatted with Mark Cuban (from Shark Tank) on his Cyber Dust. Very thoughtful and useful app. Give it a try and add me: @anindidharwar
Friday 19th September 2:41 PM.

 **Md. Armanbekh Samad**
WOW, I wish I was there:)

Saturday 20th September 12:27 PM.

 **Aaron Bird**
Not sure I want to know when I'll die, but 83 sounds okay

Shown Stories



Mevin Malikahlagh
Mevin Malikahlagh posted a video.



Saturday 20th September 12:30 PM.



Mehnoosh Sameli
WOW, I wish I was there:



Saturday 20th September 12:27 PM.



Aaron Bird
Not sure I want to know when I'll die, but 83 sounds okay



Friends View: Scaling Social Patterns



Reactions to Algorithm Outputs

❖ **Surprise, Curiosity, and Dissatisfaction**

“ it’s kind of waking up in “*the Matrix*” in a way. I mean you have what you think as your reality of like what they choose to show you. ”

(P19)

❖ **Revealing Misperceptions about Friends’ Activities**

“ I have never seen her post anything!, and I always assumed that I wasn't really that close to that person so that's fine. What the hell?! ”

(P3)

Hidden Stories

 **Wan Wan Chan**
Badminton with _____ friends. Thanks to you, I had a lot of fun! Special thanks to _____ and _____ that I've learned and improved a lot by playing with you. Hope to see you again!
Saturday 20th September 12:48 AM.

 **Shawn Lee**
Top rated video

Saturday 20th September 12:05 PM.

 **Walter Ward**
Walter Ward commented on a photo.
Friday 12th September 2:42 PM.

Shown Stories

 **Shawn Lee**
"Survival Selfie"

Saturday 20th September 9:46 AM.

 **Shawn Lee**
It took a bit longer to get to work today
Saturday 20th September 1:55 AM.

 **Shawn Lee**
_____ changed her profile picture.
I had the best days of my life with you, thank you for that, I love you..
Saturday 20th September 1:44 AM.

Content Rearrangement View

Friend Rearrangement View

Rarely Shown



Yajun Zhang



Sometimes Shown



Fawaz Daghfah



Mostly Shown



Walter Ward



Marlene Rodriguez



Fabiana Rodriguez



Mustafa Taha



Talha Ghaffar



Ali Ghaffar



Margaret Stone



Expectations of Algorithm Outputs

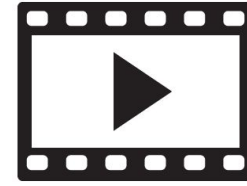
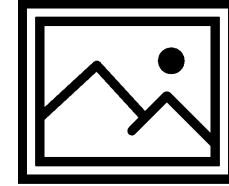
Friends



43%



Content



17%

“ A lot of what is filtered out are things that don't really pertain to me. I'm so grateful because, otherwise, it would just clutter up what I really want to see. ”

(P13)

From Algorithm Awareness to Future Behavior

Follow up: 2-6 months after the study

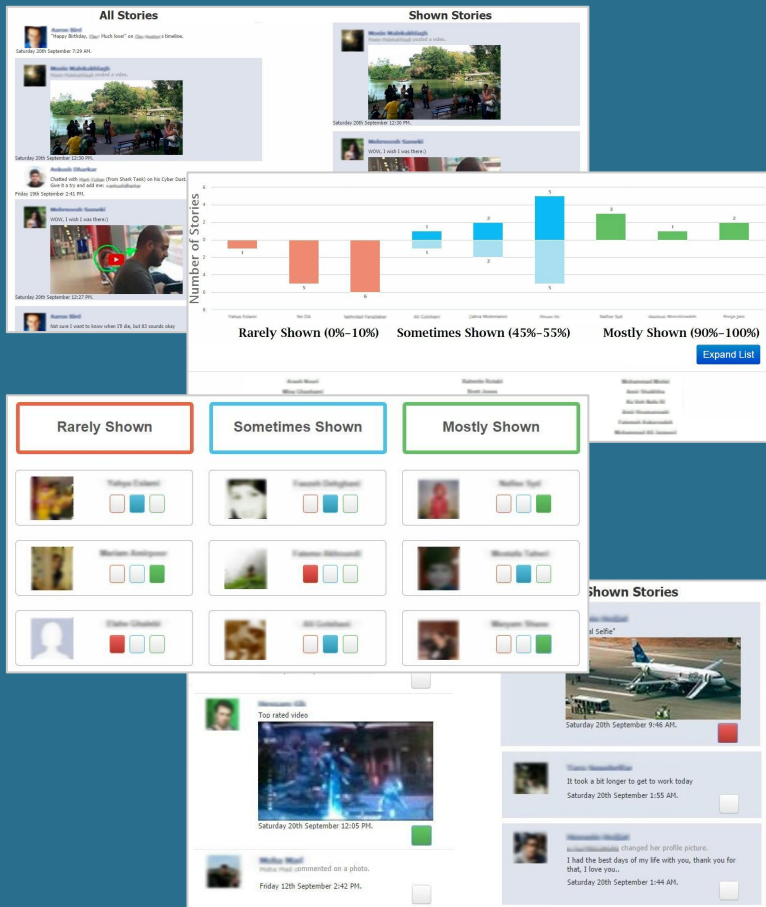
❖ **Increased Engagement**

➤ **Manipulating the Manipulation**

“ I am more selective about clicking ‘like’ because it will have consequences on what [they] see/don’t see in the future.”
(P4)

❖ **Same or Higher Satisfaction**

FeedVis



- M. Eslami, A. Rickman, K. Vaccaro, A. Aleyasen, A. Vuong, K. Karahalios, K. Hamilton, and C. Sandvig. *'I always assumed that I wasn't really that close to [her]': Reasoning about invisible algorithms in the news feed.* CHI 2015.

- Best paper award 🏆
- Top 3 Most Cited paper of CHI 2015

- M. Eslami, K. Karahalios, C. Sandvig, K. Vaccaro, A. Rickman, K. Hamilton, and A. Kirlik. *First I "like" it, then I hide it: Folk Theories of Social Feeds.* CHI 2016.

- K. Hamilton, K. Karahalios, C. Sandvig and M. Eslami. *A Path to Understanding the Effects of Algorithm Awareness.* Alt.Chi 2014.

Media Coverage: Time, The Washington Post, HuffingtonPost, BBC, CBC Radio, Fortune, Quartz, International Business Times, New Scientist, MIT Technology Review.

New Facebook tool answers the question 'Why am I seeing this post?'

[Login](#)

Catherine Shu @catherineshu / 14 hours ago

[Comment](#)

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—

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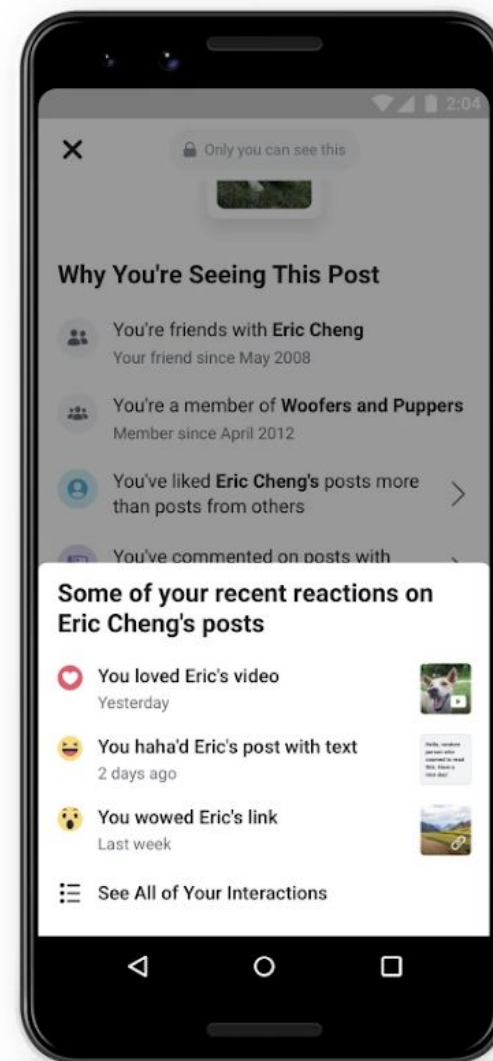
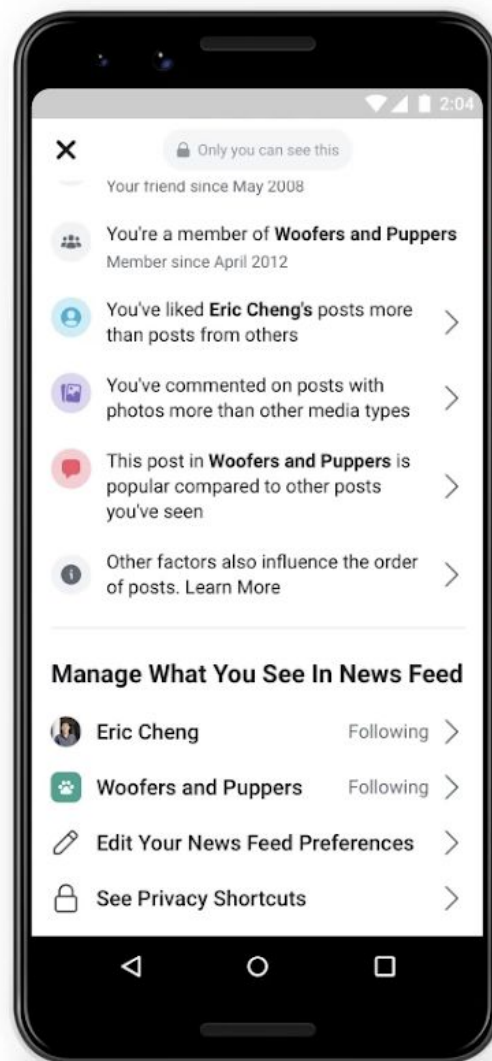
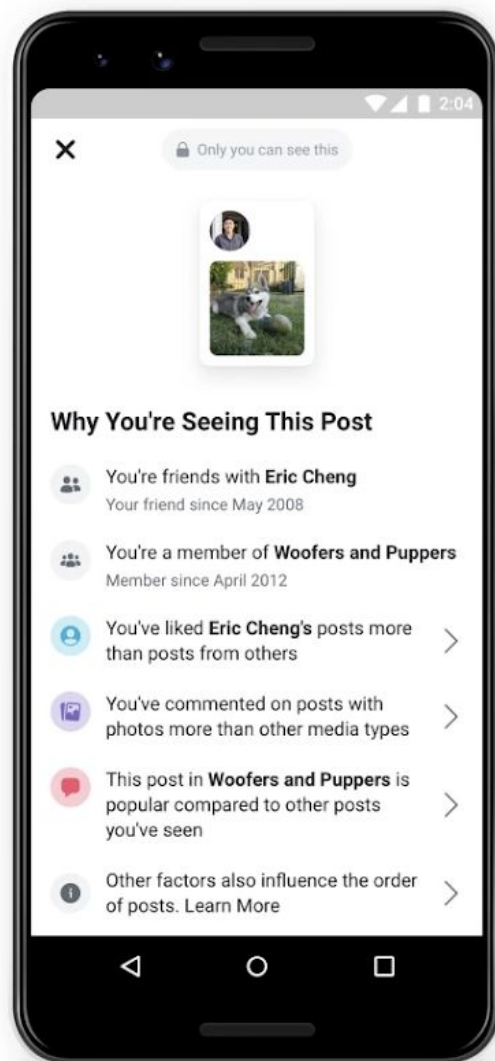
Fundings & Exits

Cybersecurity 101

Asia

Google

Search



- Transparency of Existence
- Transparency of Operation

Explainable AI Approaches

Feature Attribution

Generate feature attributions.

- SHAP
- LIME



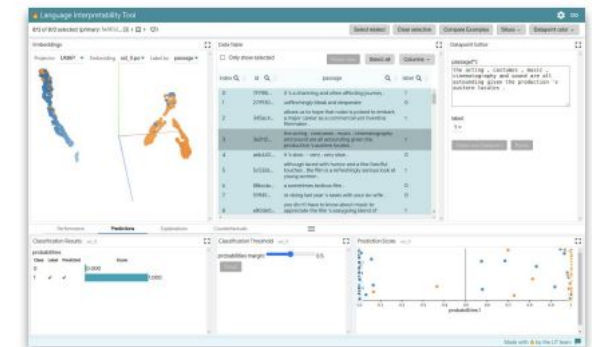
Example-based Explanation

Return similar examples to new predictions/instances



Model Analysis

Understand models performance., e.g., model cards

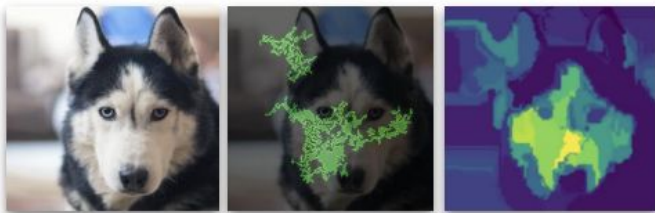


Explainable AI Approaches

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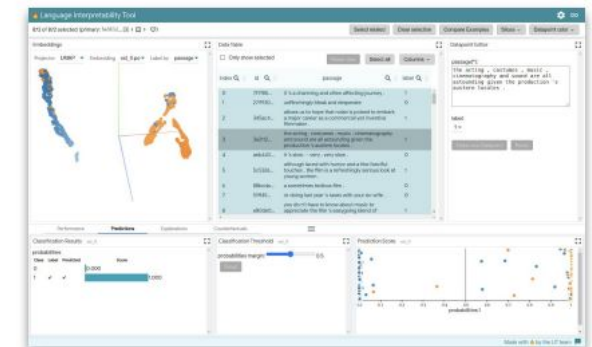
Example-based Explanation

Return similar examples to new predictions/instances



Model Analysis

Understand models performance., e.g., model cards



Feature Attributions tells you how important each input feature is

Images



Image classification

Tabular

Feature name	Feature value
start_hr	18
weekday	1
distance	1395.51
temp	16.168
dew_point	7.83396
wdsp	0
max_temp	20.7239
prcp	0.03
rain_drizzle	0
duration	11

Classification / regression

Text

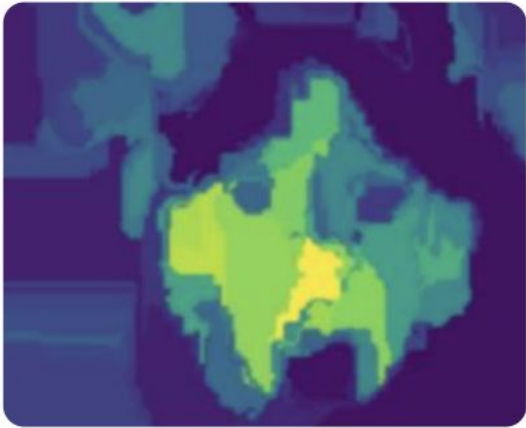
The cake tastes
delicious!

Text classification

ML task

Feature Attributions tells you how important each input feature is

Images

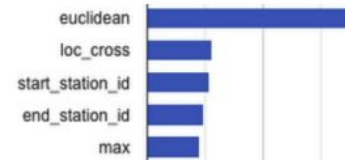


Explanations
tell you

What **image pixels** or **regions** most contributed to the classification?

Tabular

Name	Feature value	Attribution value
distance	1395.51	-2.44478
start_hr	18	-1.29039
max_temp	20.7239	0.690506
temp	16.168	0.12629
dew_point	7.83396	0.0110318
prcp	0.03	-0.00134132



How much did each **feature column** contribute to a single prediction or the model overall?

Text

The cake tastes
delicious!

Sentiment score: 0.9

How much did each **word** or **token** contribute to the text classification?

How to calculate feature Attributions?

LIME

- Ribeiro et al. 2016. **“Why Should I Trust You?” Explaining the Predictions of Any Classifier**

SHAP

- Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017).

How to calculate feature Attributions?

LIME

- Ribeiro et al. 2016. **“Why Should I Trust You?” Explaining the Predictions of Any Classifier**

SHAP

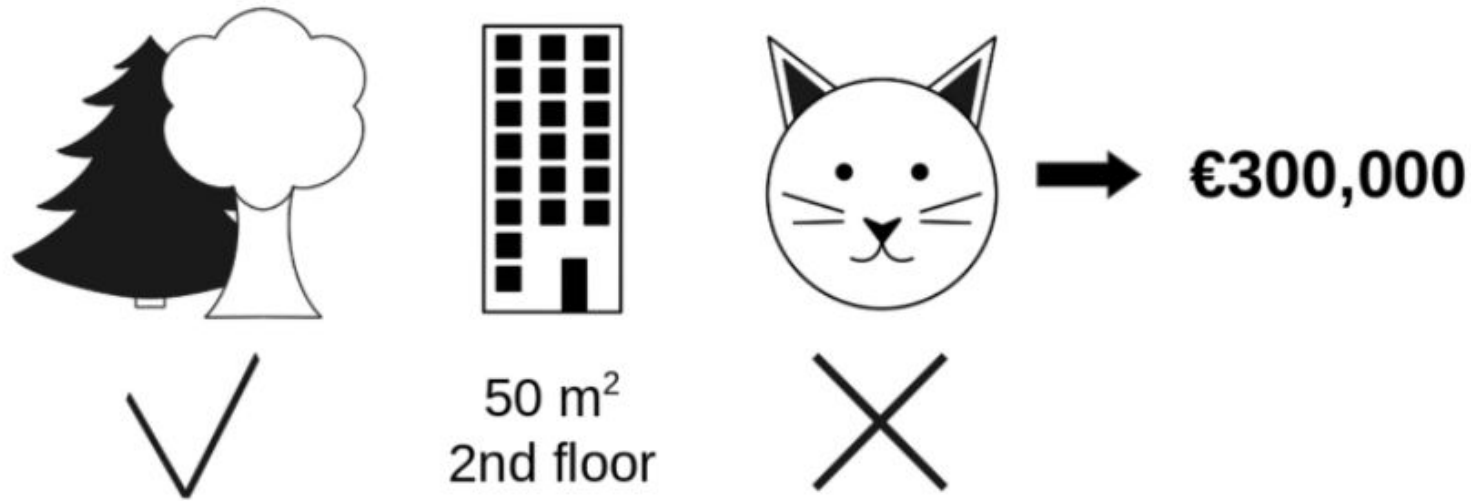
- Lundberg, Scott M., and Su-In Lee. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017).

Post hoc Explanation Techniques: SHAP



SHAP (SHapley Additive exPlanations) is a game theoretic approach to explain the output of any machine learning model. It connects optimal credit allocation with local explanations using the classic Shapley values from game theory and their related extensions (see [papers](#) for details and citations).

Post hoc Explanation Techniques: SHAP



- Example of feature contributions:
 - “Park nearby”: + €30,000
 - “Area 50”: + €10,000
 - “2nd floor”: € 0
 - “cat-banned”: - €50,000

Post hoc Explanation Techniques: SHAP

SHAP: **S**Hapley **A**dditive ex**P**lanations

- Explicit Shapley value for each feature i

$$\phi_i(f, x) = \sum_{z' \subseteq x'} \frac{|z'|!(M - |z'| - 1)!}{M!} [f_x(z') - f_x(z' \setminus i)]$$

- Computing exact Shapley values is computationally expensive
- SHAP provides a fast method for approximating these values

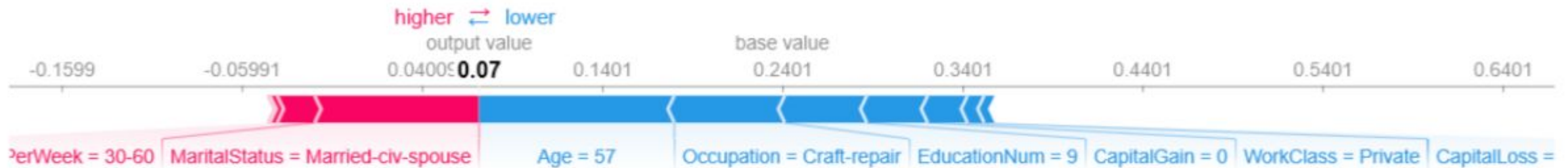
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Local Explanation Visualization (SHAP): Force Plot

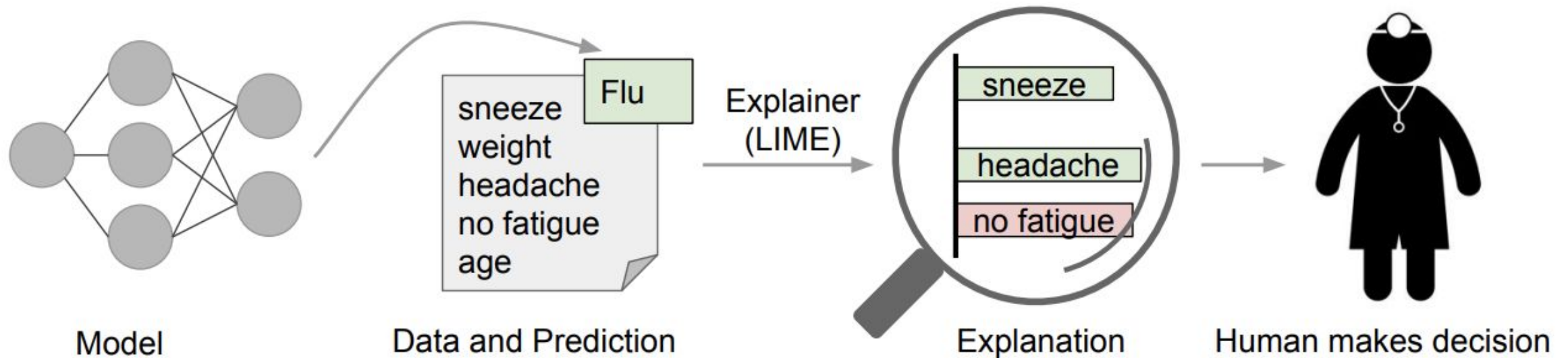
How an Individual Prediction was made



For example, if we look at the force plot above, we can see that the base value β is 0.2401. Input features such as Age (= 57), Occupation (= Craft-repair) and EducationNum (= 9) push the predicted output value to be lower, whereas input features such as MaritalStatus (= Married-civ-spouse, i.e., married to a civilian spouse) and HoursPerWeek (30-60) push the predicted output value to be higher. Ultimately, this results in the overall predicted output value of 0.07. The positive or negative scores here are SHAP values $\varphi_i(x)$ for each feature i .

LIME: “Why Should I Trust You?”

Explaining the Predictions of Any Classifier

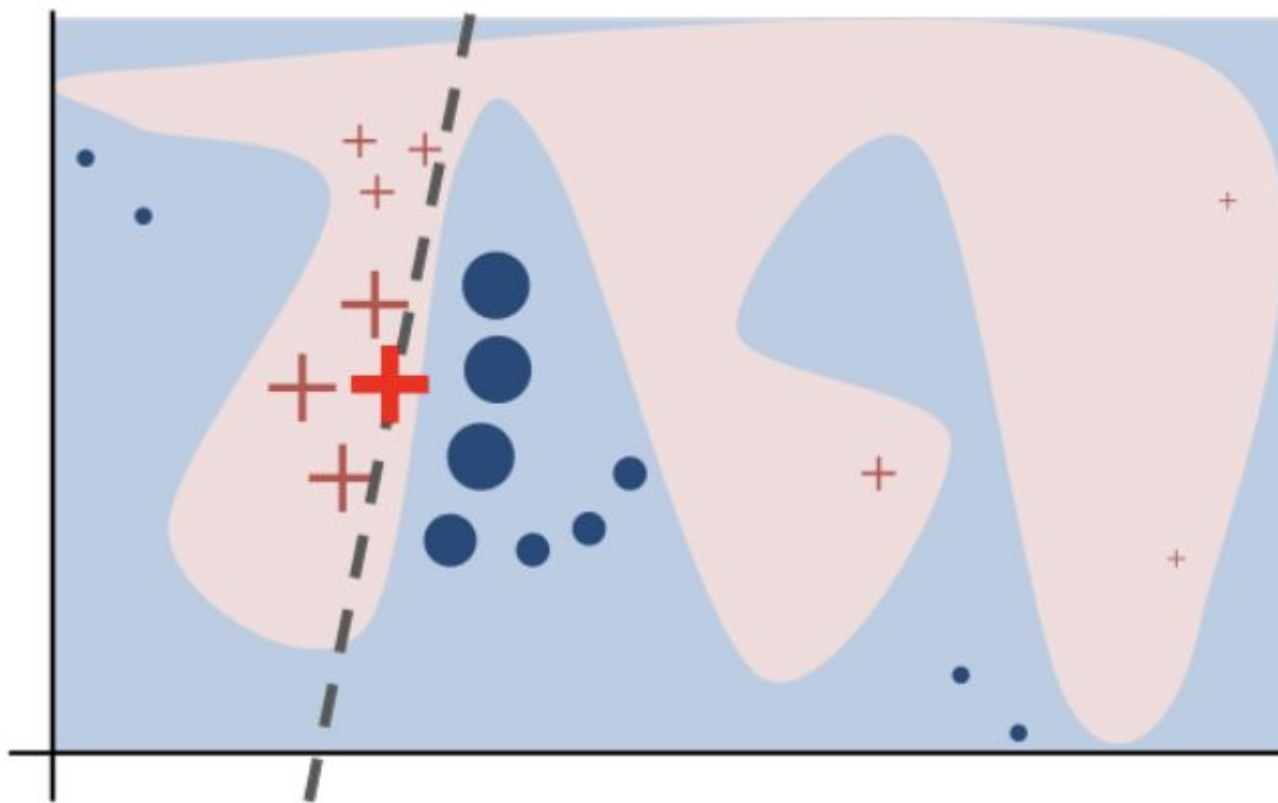


Desired Characteristics for Explainers

- **Interpretable**, i.e., provide qualitative understanding between the input variables and the response.
- **Local fidelity**
- **Model-agnostic**
- **Global perspective**

“Why Should I Trust You?”

Explaining the Predictions of Any Classifier



Example: text classification

Task: “Christianity” or “Atheism”

Example #3 of 6

True Class:  Atheism

[Instructions](#) [Previous](#) [Next](#)

Algorithm 1

Words that A1 considers important:

GOD	
mean	
anyone	
this	
Koresh	
through	

Predicted:  Atheism

Prediction correct: 

Algorithm 2

Words that A2 considers important:

Posting	
Host	
Re	
by	
in	
Nntp	

Predicted:  Atheism

Prediction correct: 

Document

From: pauld@verdix.com (Paul Durbin)
Subject: Re: DAVID CORESH IS! **GOD!**
Nntp-Posting-Host: sarge.hq.verdix.com
Organization: Verdix Corp
Lines: 8

Document

From: pauld@verdix.com (Paul Durbin)
Subject: **Re:** DAVID CORESH IS! GOD!
Nntp-Posting-Host: sarge.hq.verdix.com
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Lines: 8

Evaluation

Recruit participants from Amazon Mechanical Turk and graduate students

- 1. Can users choose which of two classifiers generalizes better?**
- 2. Based on the explanations, can users perform feature engineering to improve the model?**
- 3. Are users able to identify and describe classifier irregularities by looking at explanations?**

https://github.com/marcotcr/lime

Search or jump to... Pull requests Issues Marketplace Explore

marcotcr / lime Watch 280 Star 8k Fork 1.3k

< Code Issues 22 Pull requests 8 Actions Projects Wiki Security Insights

master 7 branches 18 tags Go to file Add file Code

marcotcr	previous fix was actually wrong	0d1a635 on Sep 10	501 commits
benchmark	removing author tag	4 years ago	
binder	adding pyDOE2 dependency	3 months ago	
doc	minor change in keras tutorial	last month	
lime	previous fix was actually wrong	last month	
.gitignore	CI: run unit tests & expand Travis configuration	4 years ago	
.travis.yml	Drop python 2 support	17 months ago	
CONTRIBUTING.md	Update CONTRIBUTING.md	4 years ago	
LICENSE	Initial commit	5 years ago	
MANIFEST.in	Add LICENSE to MANIFEST.in	4 years ago	
README.md	Update README.md	2 months ago	
citation.bib	citation	3 years ago	
setup.cfg	flake arg in setup.cfg	4 years ago	
setup.py	Merge pull request #479 from elacx/master	2 months ago	

About

Lime: Explaining the predictions of any machine learning classifier

Readme

BSD-2-Clause License

Releases 18

drop python2 support Latest on Apr 3

+ 17 releases

Packages

No packages published

Used by 777

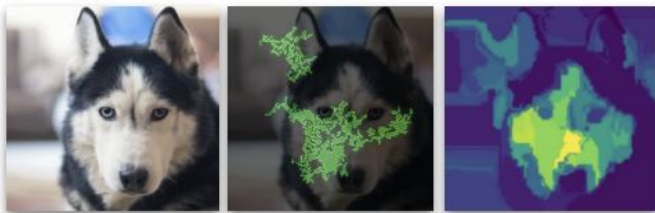
+ 769

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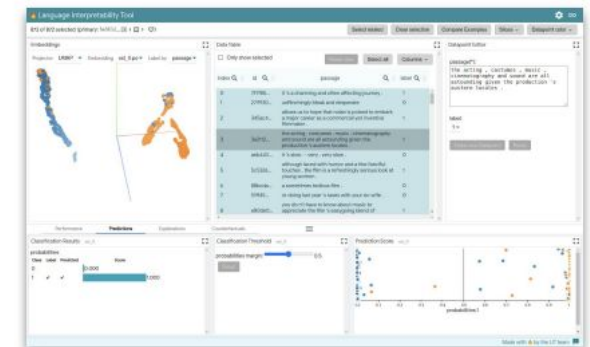
Example-based Explanation

Return similar examples to new predictions/instances



Model Analysis

Understand models performance., e.g., model cards



Explainable AI Approaches

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Generate feature attributions.

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Example-based Explanation

Return similar examples to new predictions/instances



Model Analysis

Understand models performance., e.g., model cards



Example-based explanations provides approximate nearest neighbor explanations

Images



Image classification

Tabular

start_hr	12
weekday	5
trip distance	1531.53
temp	60.8
rain	0
duration	10

Classification / regression

Text

Australian captain Ricky Ponting has left the door open for fast bowler Brett Lee to return to the side for next week's ICC Champions Trophy semi-final.

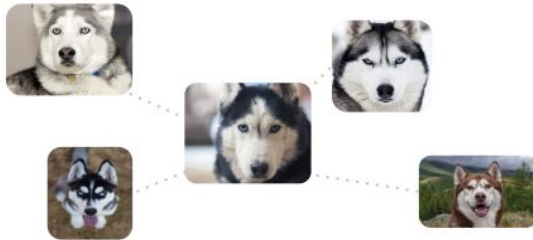
Ground truth label: WORLD
Model prediction: SPORT

Text classification

ML task

Example-based explanations provides approximate nearest neighbor explanations

Images



Explanations
tell you

What **images** in the **training dataset** are most similar to the new image?

Tabular

Start hr	Day	Distance	Temp	Rain	Duration	Distance
12	5	1531.53	60.8	0	10	0
16	5	1421.08	60.8	0	12	0.62
12	6	1596.66	50.9	0	10	1.3
14	5	2193.73	46.8	0	16	1.38
8	6	1701.77	58.8	0	11	1.69
9	5	2271.88	52.5	0	13	1.79

What **data points** in the **training dataset** are most similar to the new instance?

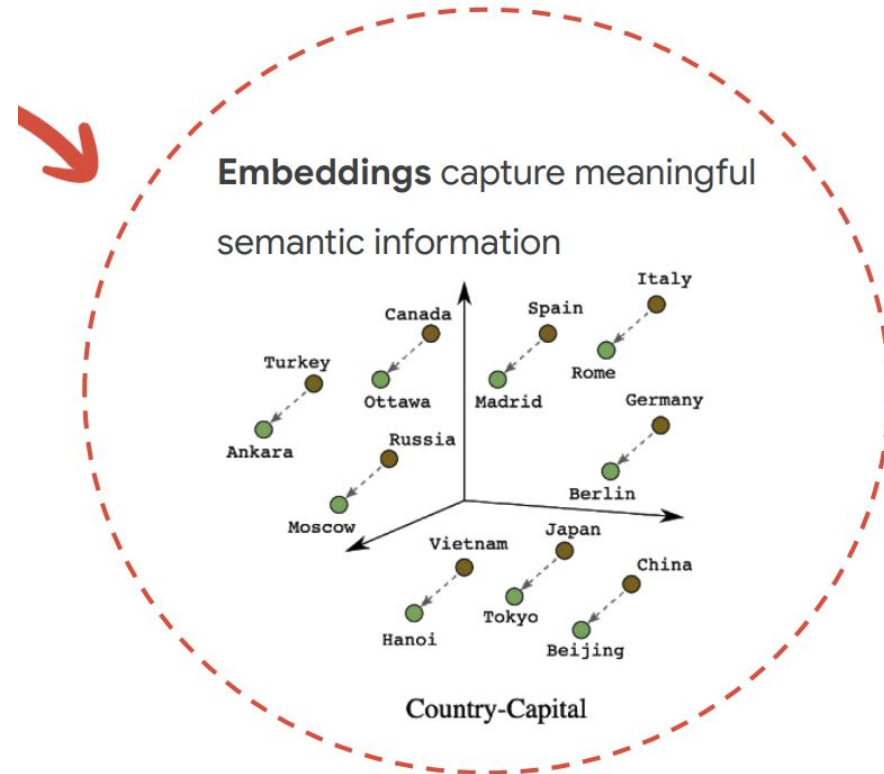
Text

- Steve Bruce paid tribute to the influence of England striker Emile Heskey after [...]. **WORLD**
- England beat Zimbabwe by eight wickets in the third one-day international at the Queens Sports Club to take an [...]. **SPORT**
- D-DAY looms for Darren Lehmann after captain Ricky Ponting was yesterday cleared to rejoin the Australian team for the [...]. **SPORT**

What **news articles** in the **training dataset** are most similar to the new article?

Steps to generate example-based explanations

1. Define similarity
2. Query for most similar examples



Explainable AI Approaches

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- LIME



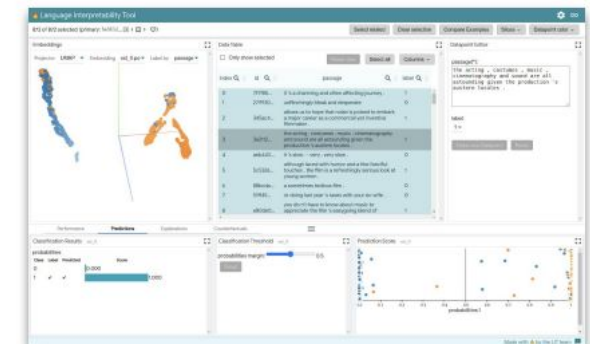
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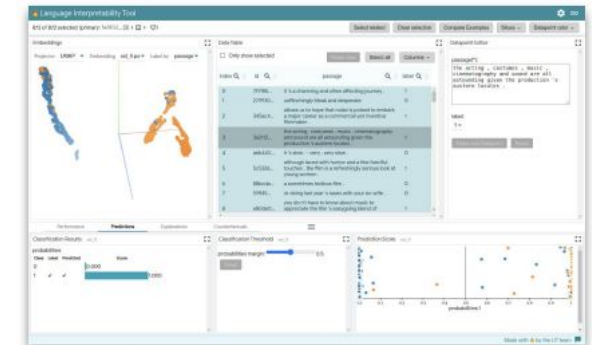
Example-based Explanation

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Model Analysis

Understand models performance., e.g., model cards



Model Number: 1

Accuracy

Over all: 69%

White American: 72%

African American: 65%

Disparity

False Positive Rate: 6%

False Negative Rate: -15%

False Positive Rate

Over all: 8%

White American: 5%

African American: 11%

False Negative Rate

Overall: 74%

White American: 83%

African American: 68%

Limitations of AI explanation techniques (feature attributions)

1. Vulnerable against adversarial attacks
2. Misused by data scientists
3. Users' trust in algorithmic decisions is not necessarily affected by the explanation (Cheng et al., 2019)

Limitations of AI explanation techniques

1. Vulnerable against adversarial attacks

LIME and SHAP are also Post hoc Explanation Techniques

- Typically consider the **complex model as a black box**
 - No internal details of the complex model required, only query access
- **Several types** of post hoc explanation techniques
 - Local vs. Global approximations, Gradient based vs. perturbation based.
- **Examples:** LIME, SHAP, Anchor, MUSE, Gradient times Input, Integrated Gradients etc.

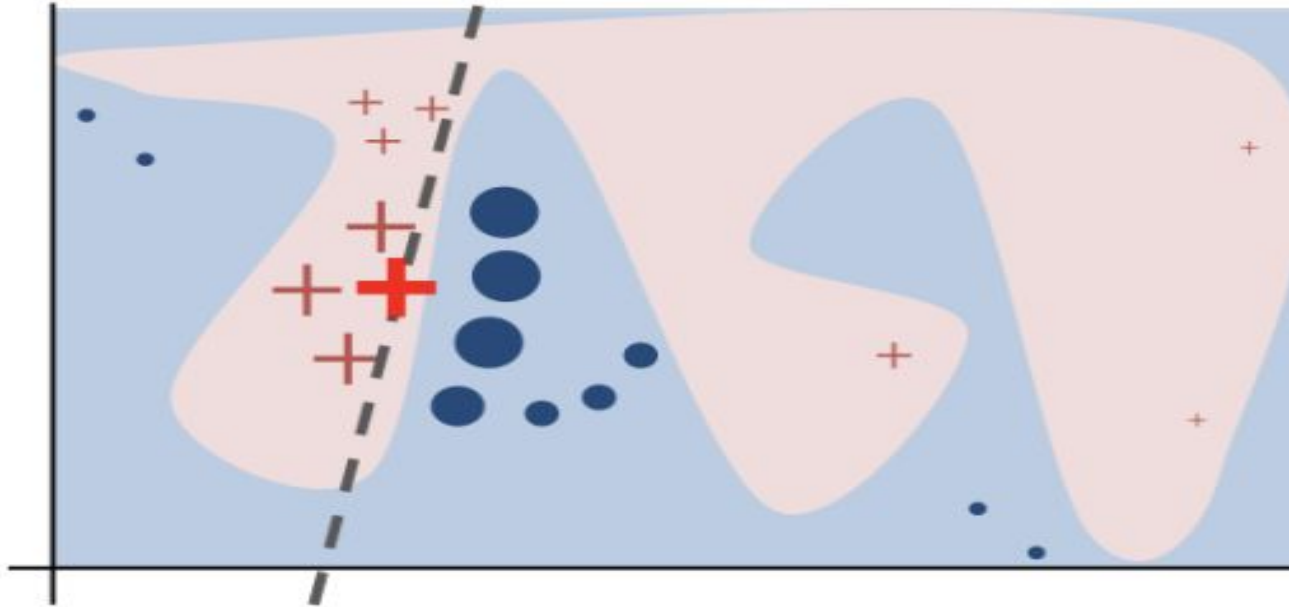
Recap: LIME & SHAP

- Model agnostic
- Local explanations
 - explain individual predictions of any classifier by learning an interpretable model (e.g., linear model) locally around each prediction
- Output feature attributions for individual instances, which capture the effect/contribution of each feature on the black box prediction

Adversarial Attacks on Post hoc Explanation Methods

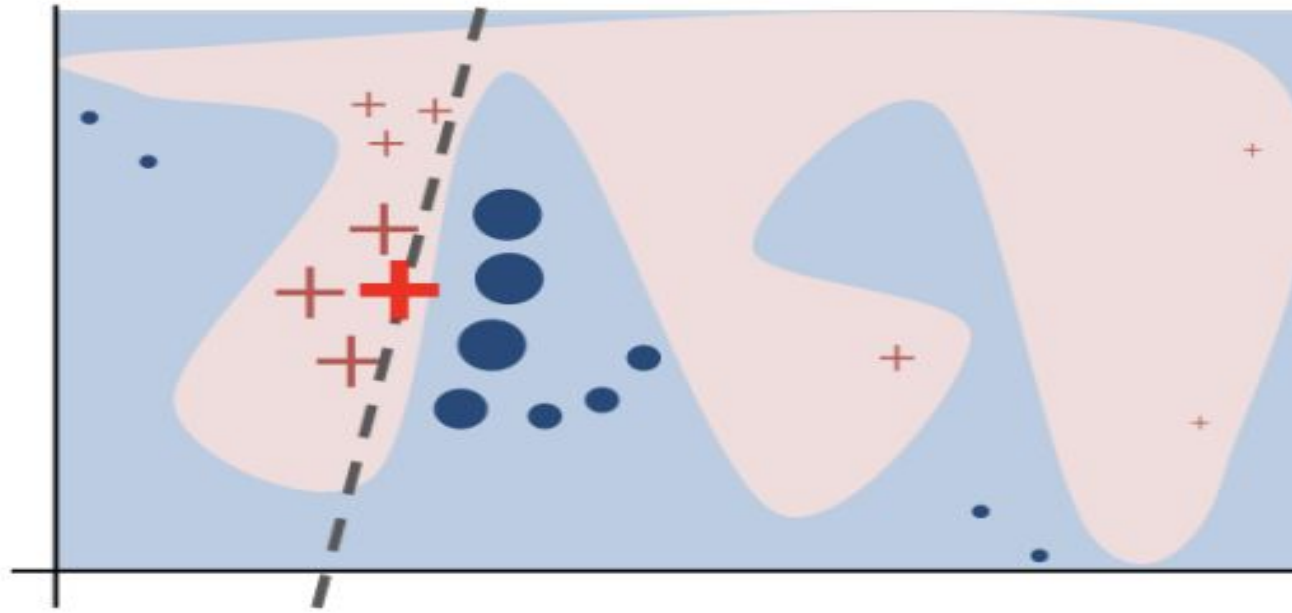
- A novel framework that can effectively mask the discriminatory biases of any black box classifier by fooling perturbation based post hoc explanation techniques
 - E.g. LIME and SHAP
- Using the proposed framework, adversaries can game perturbation based post hoc techniques to generate any arbitrary explanation of their choice.
- Experiments with real world datasets (e.g., COMPAS) reveal that our attacks are successfully fooling explanation methods.

Background: LIME & SHAP



While complex models have complex decision surfaces and are harder to explain globally, they have simpler decision boundaries in local neighborhoods

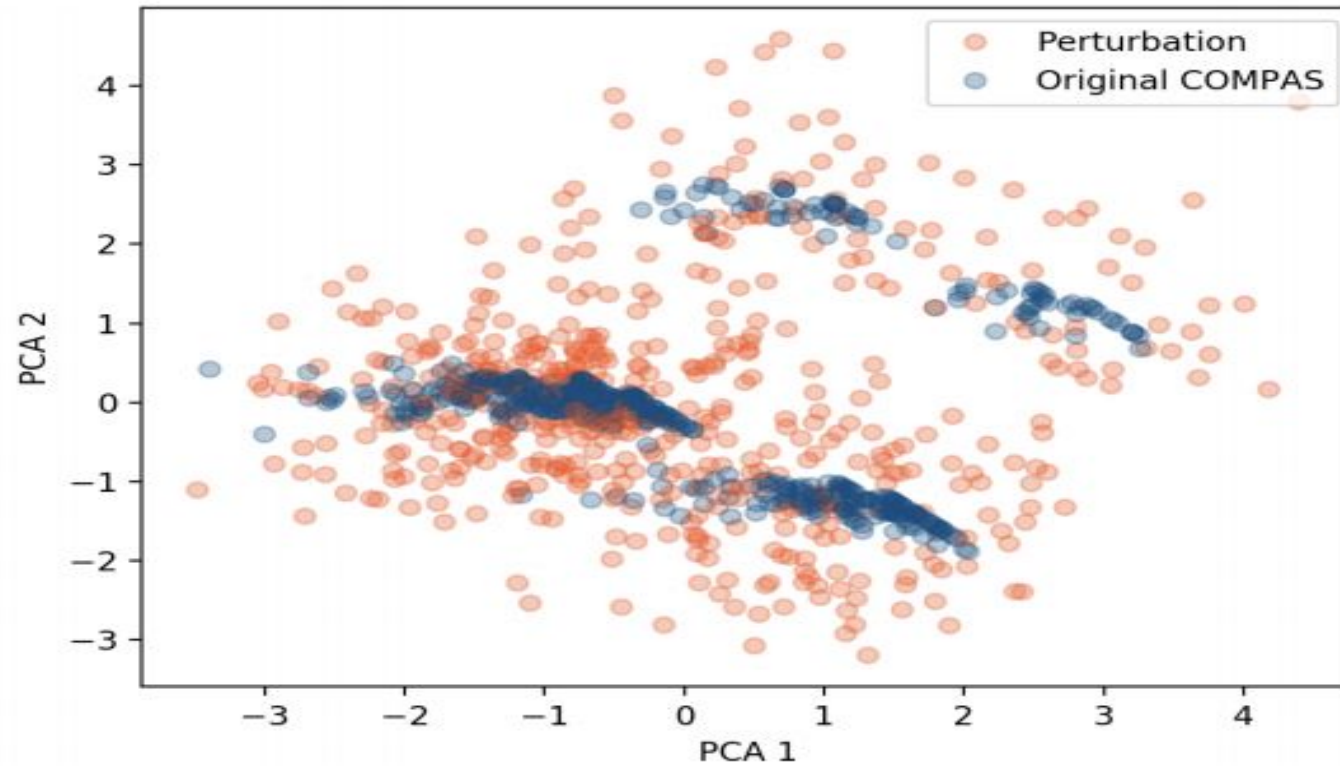
Perturbations in LIME & (Kernel) SHAP



Each data point is perturbed and a neighborhood is generated.
Local linear model is constructed by a linear regression on this neighborhood.
Class labels of data points will be the predictions of black box.

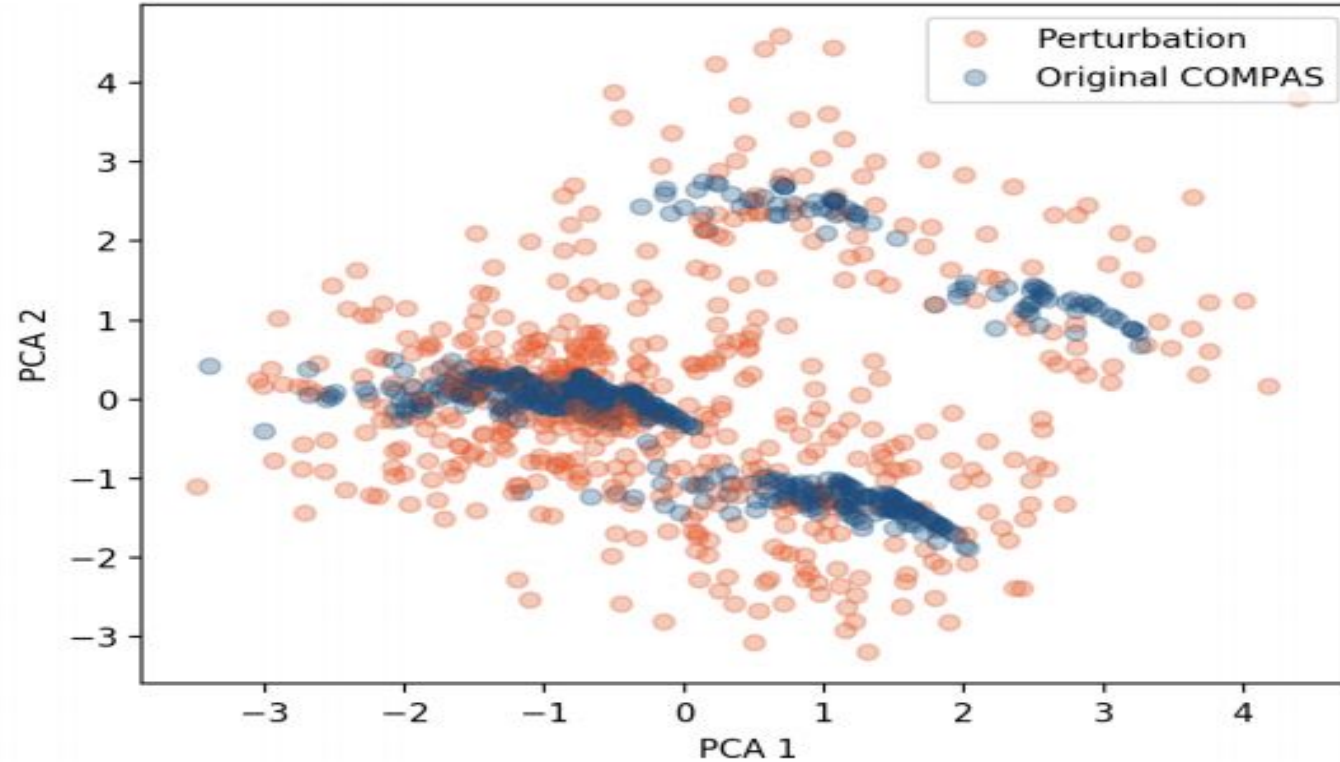
Perturbation: Out of all the features set to 1 (for a given data point), randomly sample a subset of features and set their values to 0.

Vulnerabilities of LIME/SHAP: Intuition



Several perturbed data points are out of distribution (OOD)!

Vulnerabilities of LIME/SHAP: Intuition



Adversaries can exploit this and build a classifier that is biased on in-sample data points and unbiased on OOD samples!

Limitations of AI explanation techniques

1. Vulnerable against adversarial attacks
2. Misused by data scientists

Kaur et al., CHI 2020

RESEARCH-ARTICLE • 

Interpreting Interpretability: Understanding Data Scientists' Use of Interpretability Tools for Machine Learning



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CHI '20: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems • April 2020 • Pages 1–14 • <https://doi.org/10.1145/3313831.3376219>

Questions?

In-Class Activity



In-class activity

1. **Form groups of 2**
2. **Choose an ML-based prediction system that interests you** (e.g., image prediction for radiologists, text prediction, or any other prediction tasks).
3. **Discuss how to explain the model prediction using the following approaches:**
 - a. Feature attribution,
 - b. Example-based,
 - c. Model performance.
4. **Determine which approach (or combination of approaches) seems most reasonable to use for your system and context.**
5. **Take notes** on a doc and submit them on the slack channel.

Next Session

- **Transparency and Interpretability (cont'd)**